Quiz 3 - DSC 10, Fall 2025

Full Name:

PID:

Quiz Time: ◯ 2PM ◯ 3PM ◯ 4PM ◯ 5PM

Instructions:

- This quiz consists of 4 questions. You have a total of 20 minutes to complete it.
- Please write **clearly** in the provided answer boxes; we will not grade work that appears elsewhere. Completely fill in bubbles and square boxes; if we cannot tell which option(s) you selected, you may lose points.

  ◯ A bubble means that you should only **select one choice**.

  ☐ A square box means you should **select all that apply**.
- If your answer is a string, make sure to put it in quotes. If your answer is a float, make sure to include a decimal point.
- You may use one handwritten sheet of notes. No calculators and no computers.

By signing below, you are agreeing that you will behave honestly and fairly during and after this quiz.

Signature:

Version A

Please do not open your quiz until instructed to do so.

Suppose that at the cat adoption center, all cats stay for just one day before being adopted. The `cats` DataFrame contains a **simple random sample of 500 cats** from the adoption center. In addition to each cat's `"name"` (str), we have the following information about their single day at the adoption center: number of `"treats"` consumed (int), `"nap_time"` in hours (float), and `"top_speed"` (float) in miles per hour. `cats` is sorted in **descending** order of `"top_speed"` with the first few rows shown.

	cat	treats	nap_time	top_speed
0	Tortoise	3	10.1	29.0
1	Racer	9	0.5	28.6
2	Kitkat	12	4.6	28.1
3	Tornado	5	1.7	27.9
4	Bolt	11	1.5	27.4
5	Pepper	8	2.4	27.1

Question 1

A sleepy cat is a cat that naps for at least 5 hours. You want to estimate the proportion of **all** sleepy cats that consumed at least 10 treats, based on the data in `cats`.

a) Fill in the blanks below to generate an array of 10000 bootstrapped estimates for this proportion.

```
estimates = np.array([])
subset = cats[__(a)__]
for i in np.arange(10000):
    resample = __(b)__
    proportion = __(c)__ / __(d)__
    estimates = np.append(estimates, proportion)
```

(a):

(b):

(c):

(d):

b) Fill in the blanks below so that `treats_interval` evaluates to a 90% confidence interval for the true proportion of all sleepy cats that consumed at least 10 treats.

```
ci_low = np.percentile(estimates, __(a)__)
ci_high = np.percentile(estimates, __(b)__)
treats_interval = [ci_low, ci_high]
```

(a):

(b):

c) Suppose `treats_interval` evaluates to `[0.10, 0.15]`. Select all true statements.

- ☐ Between 10% and 15% of all sleepy cats also consumed at least 10 treats.
- ☐ If we increase our confidence level to 95%, then the width of our interval will decrease and our interval will become more precise.
- ☐ There is a 90% chance that the true proportion of all sleepy cats that consumed at least 10 treats is between 10% and 15%.
- ☐ None of the above.

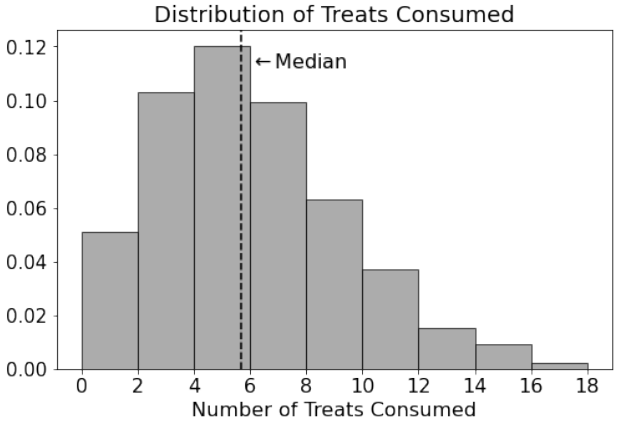
Question 2

Your friend's cat, Bugatti, is not represented in `cats`. If Bugatti's top speed is 27.2 miles per hour, what percentile is Bugatti's top speed in, relative to all the cats in the `cats` DataFrame?

◯ 1st ◯ 5th ◯ 6th ◯ 83rd ◯ 95th ◯ 99th

Question 3

The distribution of the `"treats"` column is shown below, with the median labeled.



a) Select the answer closest to the mean of `"treats"`.

◯ 4 ◯ 7 ◯ 12

b) Select the answer closest to the standard deviation of `"treats"`.

◯ 1 ◯ 4 ◯ 16

Question 4

You compute the following statistics about the `"top_speed"` column:

$$\text{mean} = 14.6 \qquad \text{median} = 16.3$$

You also know that **no cat** ran at a top speed of exactly 14.6 or 16.3 miles per hour. Select all true statements.

- ☐ Exactly half of the cats had a top speed greater than 16.3 miles per hour.
- ☐ There must be at least one cat with a top speed between 14.6 and 16.3 miles per hour.
- ☐ The output of `(cats.get("top_speed") - 16.3).sum() > 0` is `True`.
- ☐ If we were to add a row to `cats` for a cat with a top speed of 20 miles per hour, the median **must** increase.
- ☐ None of the above.

Reminders:

- **Write your PID** on the front page and on the top right corner of page 3.
- If your answer is a string, make sure to put it in **quotes**.
- If your answer is a float, make sure to include a **decimal point**.
- Stay in your seat until the quiz is over. No leaving early.